LITERATURE SURVEY

Interactive AR-Based Brain Tumour Visualization and Tumour-Progression Simulation

Medical Imaging, Radiomics, Machine Learning, Treatment-Response Modelling, and Patient-Specific Simulation

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Abstract

Background—Brain tumours are spatially heterogeneous lesions whose enhancing tissue, infiltrative margin, necrosis, oedema and relationship to eloquent anatomy are difficult to infer from isolated two-dimensional slices. Objective—This survey evaluates how augmented and virtual reality, multimodal MRI segmentation, radiomics, machine learning, mathematical growth models and treatment-response simulation can be combined into an interactive, scientifically defensible educational system. Search approach—A PRISMA-inspired search was performed on 19 July 2026 using Consensus and Scite alongside PubMed/DOI and publisher metadata checks. Searches covered 2000–2026, prioritised 2015–2026, and selected exactly 30 unique peer-reviewed papers: six AR/VR, six segmentation/3D reconstruction, five radiomics, five prediction, four growth-modelling, and four treatment-planning or patient-specific simulation papers. Major findings—Automatic multimodal MRI segmentation and 3D visualization are mature enough for a student prototype; nnU-Net-type pipelines provide a practical baseline, while PyRadiomics supports transparent feature extraction. AR can improve spatial understanding and workflow, but registration, occlusion, depth perception and human factors remain limiting. Prognostic classifiers, future-mask predictors and mechanistic growth simulators address different endpoints and cannot be substituted for one another. Research gaps—Evidence is constrained by small retrospective cohorts, scanner domain shift, weak external validation, scarce longitudinal treatment data, limited uncertainty visualization and few integrated AR-plus-growth systems. Conclusion—A defensible prototype should combine validated segmentation, mesh-based mobile AR and clearly labelled educational scenarios. Any progression output must be probabilistic or mechanistic, visibly uncertain, clinician-supervised and explicitly non-diagnostic. Clinical forecasting and treatment prescription remain outside the evidence supported by this corpus.

Keywords

augmented reality; virtual reality; glioma; glioblastoma; multimodal MRI; tumour segmentation; radiomics; machine learning; reaction–diffusion model; treatment response; digital twin; uncertainty visualization.

1. Introduction

Brain tumours comprise biologically diverse primary and metastatic neoplasms arising within or affecting the central nervous system. Gliomas are especially relevant to the proposed application because their imaging phenotype often contains enhancing tumour, non-enhancing or necrotic core, and surrounding FLAIR abnormality that may reflect oedema and infiltrative disease. Location matters as much as volume: a small lesion near motor, language, visual or memory networks may be functionally more consequential than a larger lesion in less eloquent tissue. Shape is irregular, infiltration may extend beyond visible enhancement, and surgery or treatment can alter anatomy. These properties make a brain tumour a dynamic spatial process rather than a simple spherical object [8], [15], [23]–[26].

MRI is the principal soft-tissue imaging source for glioma visualization. T1-weighted MRI provides anatomy; post-contrast T1 highlights blood–brain-barrier disruption and enhancement; T2 and FLAIR emphasize water-rich abnormal tissue. CT contributes bone, calcification and acute haemorrhage information, while PET may contribute metabolism or hypoxia, as in patient-specific radiation modelling [25]. Multimodal fusion is therefore scientifically attractive but operationally difficult: images must be de-identified, co-registered, resampled, bias-corrected and normalized before labels or features can be compared [8], [12], [16].

Ordinary two-dimensional viewing requires the observer to integrate many slices and mentally transform image coordinates into three-dimensional anatomy. Mixed-reality training was developed specifically to reduce this spatial and cognitive burden [1]. AR overlays virtual objects on the real environment; VR replaces the environment with an immersive one; mixed reality supports spatially anchored interaction. Their value is strongest when a segmented lesion, vessels and functional anatomy can be explored from arbitrary viewpoints. Their limitations are equally concrete: misregistration, occlusion, limited field of view, depth ambiguity, display discomfort, attention capture and motion sickness can turn an apparently compelling visualization into a misleading one [2], [4], [5].

Progression simulation is more difficult than visualization because “future tumour” has several meanings. A classifier can estimate the probability of recurrence; a survival model estimates time-to-event risk; a longitudinal network can forecast a future mask; a reaction–diffusion equation can simulate cell-density evolution; and a generative model can synthesize a plausible image. These outputs differ in target, evidence and validation. A treatment module adds further causal difficulty because surgery, radiation and chemotherapy responses depend on dose, extent of resection, molecular status, pharmacology and patient condition [19], [21], [22], [25]–[27].

The proposed application is intended primarily for education, visualization, research support and treatment explanation. It is not an autonomous diagnostic system and must not prescribe treatment. Simulated trajectories should be labelled as scenarios, not forecasts of certainty; interfaces should show assumptions, confidence or parameter ranges, and the source of each output. For a student project, scientific credibility comes from restricting claims to what has been validated, retaining human oversight, and separating an educational animation from a patient-specific research prediction.

2. Literature-Search Methodology

A reproducible, PRISMA-inspired search was completed on 19 July 2026. Consensus was used to discover peer-reviewed candidates and retrieve bibliographic records; Scite was used to cross-check metadata, available full-text excerpts, citation context and editorial notices. PubMed/PMC, IEEE Xplore, SpringerLink, ScienceDirect, Nature Portfolio, Wiley, Oxford Academic, Frontiers, MDPI, ACM-indexed proceedings, MICCAI proceedings and the International Journal of Computer Assisted Radiology and Surgery were treated as eligible sources. DOI landing pages and stable publisher records were used for final traceability. This report is a structured literature survey, not a registered systematic review.

The date range was 2000–2026, with priority given to 2015–2026. Older papers were admitted only when foundational: the U-Net architecture, core radiomics methods and proliferation–invasion modelling. Search strings included: (“augmented reality” AND “brain tumour”); (“virtual reality” AND “neurosurgery”); (“mixed reality” AND “brain tumour visualization”); (“brain MRI” AND “3D reconstruction”); (“glioma segmentation” AND “deep learning”); (“BraTS” AND “tumour segmentation”); (“brain tumour radiomics”); (“glioblastoma” AND “radiomics” AND “survival”); (“tumour progression prediction” AND “MRI”); (“glioma recurrence prediction”); (“treatment response prediction” AND “brain tumour”); (“reaction diffusion model” AND “glioma”); (“patient-specific tumour simulation”); (“digital twin” AND “oncology”); (“AR surgical navigation” AND “neurosurgery”); and (“radiotherapy planning” AND “brain tumour”).

Inclusion criteria were: peer-reviewed journal or full conference paper; direct relevance to at least one architecture component; traceable title, authorship, year, venue and DOI or stable link; original empirical, computational, data-descriptor or systematic-review evidence; and sufficient methodological information for critical interpretation. Exclusion criteria were: blogs, news, vendor claims, dissertations, student reports, unpublished presentations, duplicate records, preprints without a peer-reviewed version, papers unrelated to intracranial imaging or transferable methodology, and studies whose bibliographic identity could not be verified. Reviews were retained where they synthesized device or modelling evidence, but were labelled as reviews rather than primary clinical validation.

2.1 PRISMA-Inspired Selection Summary

Identification yielded a broad candidate pool from the connector searches and the supplied PDF. Duplicate titles and DOI variants were removed. Titles and abstracts were screened for relevance to visualization, segmentation, radiomics, prediction, spatial growth or treatment simulation. Full metadata were then assessed for peer-review status and traceability. The supplied seven-page PDF contributed 20 candidate records but included arXiv challenge papers and did not satisfy the exact-30 rule. After replacing non-peer-reviewed or insufficiently relevant records and balancing categories, 30 unique papers were included. Common exclusion reasons were preprint-only status, non-brain applications without essential methodological transfer, absence of a stable bibliographic record, purely commercial description, and claims unsupported by a patient, dataset, simulation or user evaluation.

Evidence strength was rated informally. “Strong” indicates a large or multi-centre dataset, systematic review, established benchmark/data resource, external validation, or high reproducibility. “Moderate” indicates a

credible original study with useful validation but retrospective, single-centre, simulation or transferability constraints. “Limited” indicates a small pilot, very small longitudinal cohort, perspective, or feasibility system without independent clinical validation. Ratings describe evidentiary support for this project—not overall scientific quality.

3. Summary Table of Exactly 30 Papers

No.	Full title	Authors	Year	Venue	Category	Objective	Dataset/sample	Modality	Method/model	AR/VR HW/SW	Major findings	Metrics	Limitations	Project relevance	DOI	PMID / stable link
1	Training for Planning Tumour Resection: Augmented Reality and Human Factors	K. Abhari et al.	2020	IEEE Trans. Biomed. Eng.	AR/VR	Evaluate mixed AR/VR resection-planning training	Clinicians and non-clinicians; exact subgroup totals reported in paper	Preoperative MRI	Mixed-reality planning and human-factors study	Custom mixed AR/VR system	Improved non-clinician performance and reduced clinician task time	Task performance; completion time	Training task; no patient-outcome trial	Direct evidence for spatial training	10.1109/TBME.2014.2385874	https://doi.org/10.1109/TBME.2014.2385874
2	Augmented reality in neurosurgery: a systematic review	A. Meola et al.	2021	Neurosurg. Rev.	AR/VR	Review in-vivo neurosurgical AR systems	18 studies; 195 lesions	Multiple	Systematic review of tracking, registration and displays	Microscope, cameras, HMDs	Clinical feasibility across subspecialties; limited clinical evidence	Qualitative synthesis	Heterogeneous, mostly small studies	Defines registration and display constraints	10.1007/s10143-016-0732-9	https://doi.org/10.1007/s10143-016-0732-9
3	Augmented reality head-mounted display-based incision planning in cranial neurosurgery: a prospective pilot study	M. E. Ivan et al.	2021	Neurosurg. Focus	AR/VR	Test HMD incision planning	11 cranial cases	CT/MRI	Prospective feasibility comparison	Microsoft HoloLens	Feasible in all cases; correspondence good/adequate in 10/11	Qualitative correspondence; feasibility	Small pilot; registration optimization required	Supports patient-specific AR overlay	10.3171/2021.5.FOCUS20735	https://doi.org/10.3171/2021.5.FOCUS20735
4	Augmented reality visualization in brain lesions: a prospective randomized controlled evaluation of its potential and current limitations in navigated microneurosurgery	A. L. Roethe et al.	2022	Acta Neurochir.	AR/VR	Compare AR microscope visualization with conventional navigation	55 intracranial lesions	MRI	Prospective randomized evaluation	AR-navigated microscope	Reduced pointer-navigation effort; occlusion, depth and accuracy remained concerns	Use time; pointer checks; usability ratings	Single clinical setting; visualization limitations	Best clinical evidence for AR benefits and hazards	10.1007/s00701-021-05047-x	https://doi.org/10.1007/s00701-021-05047-x
5	Utility of optical see-through head mounted displays in augmented reality-assisted surgery: a systematic review	M. Birlo et al.	2022	Med. Image Anal.	AR/VR	Assess OST-HMD surgical evidence	91 articles	Multiple	Systematic review	HoloLens and other OST-HMDs	Most evidence was phantom/setup or feasibility; registration and human factors dominate	Study-design and system characteristics	Heterogeneity; limited outcome evidence	Frames evidence strength for HMD deployment	10.1016/j.media.2022.102361	https://doi.org/10.1016/j.media.2022.102361
6	HoloBrain: 3D low-cost mobile augmented reality rendering of brain tumour using the GVF snake model segmentation	M. A. Guerroudji et al.	2024	Comput. Methods Biomech. Biomed. Eng. Imaging Vis.	AR/VR	Develop low-cost mobile AR tumour rendering	Phantom/sample MRI evaluation; exact clinical n not reported	MRI	GVF snake segmentation, 3D rendering and AR	Smartphone AR	Demonstrated accessible 3D tumour visualization; millimetric phantom errors reported	Registration/position error	Controlled setting; not clinical forecasting	Closest match to student mobile-AR goal	10.1080/21681163.2024.2302051	https://doi.org/10.1080/21681163.2024.2302051
7	U-Net: Convolutional Networks for Biomedical Image Segmentation	O. Ronneberger et al.	2021	MICCAI	Segmentation	Introduce encoder-decoder segmentation network	30 annotated microscopy images plus augmentation	Microscopy (not brain MRI)	U-shaped CNN with skip connections	Not applicable	Established highly reusable biomedical segmentation architecture	Overlap/error measures	Not a brain-tumour study; 2D original formulation	Foundational architecture for tumour masks	10.1007/978-3-319-24574-4_28	https://doi.org/10.1007/978-3-319-24574-4_28
8	Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features	S. Bakas et al.	2023	Sci. Data	Segmentation	Release expert labels and radiomic descriptors	243 TCGA glioma subjects	T1, T1CE, T2, FLAIR	Expert-approved labels plus feature extraction	Not applicable	Provides reusable labelled multimodal MRI resource	Data descriptor quality controls	Retrospective heterogeneous archive	Supports training and patient-specific mesh creation	10.1038/sdata.2017.117	https://doi.org/10.1038/sdata.2017.117
9	Brain tumor segmentation with DeepMedic	K. Kamnitsas et al.	2019	Med. Image Anal.	Segmentation	Segment brain lesions with multiscale 3D CNN	BRATS 2015 and other lesion datasets	Multimodal MRI	Dual-pathway 3D CNN with CRF	Not applicable	Competitive lesion segmentation and efficient multiscale context	Dice and challenge metrics	Benchmark-centric; compute intensive	Supports automatic volumetric masks	10.1016/j.media.2016.10.004	https://doi.org/10.1016/j.media.2016.10.004
10	3D MRI brain tumor segmentation using autoencoder regularization	A. Myronenko	2019	BrainLes/MICCAI	Segmentation	Improve 3D segmentation with auxiliary reconstruction	BraTS 2018	T1, T1CE, T2, FLAIR	Asymmetric 3D encoder-decoder with variational autoencoder branch	Not applicable	Winning BraTS 2018 approach; strong subregion segmentation	Dice; Hausdorff distance	Challenge data; memory demand	High-quality masks for visualization	10.1007/978-3-030-11726-9_28	https://doi.org/10.1007/978-3-030-11726-9_28
11	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation	F. Isensee et al.	2021	Nat. Methods	Segmentation	Automate architecture and pipeline configuration	23 public biomedical datasets	Multiple, including MRI	Self-configuring 2D/3D U-Net framework	Not applicable	State-of-the-art performance without manual dataset-specific redesign	Challenge metrics across datasets	Compute and training cost; not uncertainty-aware by default	Practical baseline for student segmentation	10.1038/s41592-020-01008-z	https://doi.org/10.1038/s41592-020-01008-z
12	Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images	R. Ranjbarzadeh et al.	2021	Sci. Rep.	Segmentation	Improve multimodal glioma subregion segmentation	BraTS 2018	T1, T1CE, T2, FLAIR	Cascade CNN with attention	Not applicable	Reported high Dice for whole, core and enhancing tumour	Dice; sensitivity; specificity; Hausdorff-type measures	Single benchmark; cross-site generalization uncertain	Provides a reproducible segmentation option	10.1038/s41598-021-90428-8	https://doi.org/10.1038/s41598-021-90428-8
13	Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach	H. J. W. L. Aerts et al.	2018	Nat. Commun.	Radiomics	Demonstrate quantitative imaging phenotypes and prognosis	1,019 lung/head-and-neck cancer patients	CT	Handcrafted radiomics, stability testing and survival modelling	Not applicable	Showed transferable prognostic imaging signatures	Concordance index; validation statistics	Not brain MRI; retrospective	Foundational radiomics methodology	10.1038/ncomms5006	https://doi.org/10.1038/ncomms5006
14	Computational radiomics system to decode the radiographic phenotype	J. J. M. van Griethuysen et al.	2019	Cancer Res.	Radiomics	Provide standardized open-source radiomics extraction	Reference datasets and phantoms	CT/MRI/PET supported	PyRadiomics feature extraction and validation	PyRadiomics	Reproducible implementation of first-order, shape and texture features	Feature agreement and validation tests	Reproducibility still depends on acquisition/segmentation	Core feature-extraction tool	10.1158/0008-5472.CAN-17-0339	https://doi.org/10.1158/0008-5472.CAN-17-0339
15	Radiomic profiling of glioblastoma: identifying an imaging predictor of patient survival with improved performance over established clinical and radiologic risk models	P. Kickingereder et al.	2018	Radiology	Radiomics	Test MRI radiomics for survival stratification	181 discovery and 79 validation patients	Multiparametric MRI	Radiomic signature and multivariable survival modelling	Not applicable	Radiomics improved prognostic modelling beyond conventional factors	C-index; survival analyses	Retrospective; feature stability/domain shift	Evidence for MRI-derived prognosis features	10.1148/radiol.2016160845	https://doi.org/10.1148/radiol.2016160845
16	Standardization of brain MR images across machines and protocols: bridging the gap for MRI-based radiomics	A. Carré et al.	2020	Sci. Rep.	Radiomics	Quantify normalization effects on feature robustness	Multiscanner glioma MRI cohort	MRI	Intensity normalization and discretization comparison	PyRadiomics-compatible pipeline	Normalization improved robustness for many feature families	Feature robustness; classification performance	No universal setting across all features/scanners	Justifies strict preprocessing and harmonization	10.1038/s41598-020-69298-z	https://doi.org/10.1038/s41598-020-69298-z
17	Multiparametric MR radiomics in brain glioma: models comparation to predict biomarker status	J. He et al.	2022	BMC Med. Imaging	Radiomics	Predict IDH, MGMT and TERT status	81 glioma patients	Multiparametric MRI	107 features/sequence, LASSO and classification	PyRadiomics	Combined clinical-radiomics models performed best for several biomarkers	ROC AUC	Small single-centre cohort; internal validation	Demonstrates interpretable biomarker module	10.1186/s12880-022-00865-8	https://doi.org/10.1186/s12880-022-00865-8
18	A deep learning-based radiomics model for prediction of survival in glioblastoma multiforme	H. Lao et al.	2017	Sci. Rep.	ML prediction	Predict overall survival from MRI features	112 GBM patients	MRI	Deep learned imaging features plus survival model	Not applicable	Imaging features showed prognostic potential	C-index; survival stratification	Retrospective; limited sample/external validation	Supports probability/risk layer, not spatial simulation	10.1038/s41598-017-10649-8	https://doi.org/10.1038/s41598-017-10649-8
19	Longitudinal brain tumor segmentation prediction in MRI using feature and label fusion	L. Pei et al.	2020	Biomed. Signal Process. Control	ML prediction	Predict future tumour segmentation from longitudinal MRI	9 BraTS 2015 longitudinal patients	Longitudinal MRI	Feature/label fusion with reaction-diffusion-derived density feature	Not applicable	Demonstrated future-mask prediction feasibility	Spatial overlap and segmentation errors	Extremely small cohort; research proof-of-concept	Directly informs timeline forecasting distinction	10.1016/j.bspc.2019.101648	https://doi.org/10.1016/j.bspc.2019.101648
20	Development of prediction models for one-year brain tumour survival using machine learning: a comparison of accuracy and interpretability	C. Charlton et al.	2023	Comput. Methods Programs Biomed.	ML prediction	Compare interpretable survival classifiers	1,028 brain-tumour patients	Clinical records; imaging not central	Interpretable and black-box ML comparison	Not applicable	Interpretable models were competitive; clinical variables were influential	AUC/accuracy and interpretability analyses	Single health-system retrospective data	Supports explainable clinical-data integration	10.1016/j.cmpb.2023.107482	https://doi.org/10.1016/j.cmpb.2023.107482
21	Estimating Progression-Free Survival in Patients with Primary High-Grade Glioma Using Machine Learning	A. Kwiatkowska-Miernik et al.	2024	J. Clin. Med.	ML prediction	Estimate PFS after recurrence/treatment	51 recurrent high-grade glioma patients	MRI plus clinical data	Radiomics, random forest and survival modelling	Not applicable	High internal performance reported	C-index; 1-MAPE	Small retrospective cohort; no external validation	Example of probabilistic PFS estimation	10.3390/jcm13206172	https://doi.org/10.3390/jcm13206172
22	Prediction of Pseudoprogression versus Progression using Machine Learning Algorithm in Glioblastoma	B.-S. Jang et al.	2018	Sci. Rep.	ML prediction	Classify progression versus pseudoprogression	Post-chemoradiotherapy GBM cohort; exact n reported in paper	Conventional post-treatment MRI	Machine-learning classification	Not applicable	Demonstrated feasibility of automated pseudoprogression discrimination	AUC; sensitivity; specificity	Retrospective; endpoint and imaging heterogeneity	Treatment-response state classifier, not growth model	10.1038/s41598-018-31007-2	https://doi.org/10.1038/s41598-018-31007-2
23	A quantitative model for differential motility of gliomas in grey and white matter	K. R. Swanson et al.	2000	Cell Prolif.	Growth modelling	Model glioma proliferation and tissue-dependent invasion	Mathematical/simulation study	MRI-informed conceptual model	Reaction-diffusion proliferation-invasion model	Not applicable	Established differential migration in grey/white matter modelling	Simulation fit/qualitative agreement	Simplified biology and boundary assumptions	Foundational spatial growth engine	10.1046/j.1365-2184.2000.00177.x	https://doi.org/10.1046/j.1365-2184.2000.00177.x
24	The evolution of mathematical modeling of glioma proliferation and invasion	H. L. P. Harpold et al.	2007	J. Neuropathol. Exp. Neurol.	Growth modelling	Synthesize proliferation-invasion modelling foundations	Review of mathematical and clinical studies	MRI and pathology literature	Reaction-diffusion model review	Not applicable	Clarified model parameters, infiltration and clinical interpretation	Not applicable	Model assumptions exceed measurable patient information	Explains mechanistic versus ML simulation	10.1097/01.jnen.0000245080.64495.7e	https://doi.org/10.1097/01.jnen.0000245080.64495.7e
25	A patient-specific computational model of hypoxia-modulated radiation resistance in glioblastoma using 18F-FMISO-PET	R. C. Rockne et al.	2015	J. R. Soc. Interface	Growth modelling	Model patient-specific growth and radiation resistance	Patient-specific imaging cohort; exact n reported in paper	MRI and FMISO-PET	Reaction-diffusion plus hypoxia/radiotherapy model	Not applicable	Linked spatial hypoxia information with simulated radiation response	Model-data fit and survival/response associations	Small cohort; parameter identifiability	Advanced research treatment-scenario basis	10.1098/rsif.2014.1174	https://doi.org/10.1098/rsif.2014.1174
26	Personalized radiotherapy design for glioblastoma using mathematical tumor models, genomics and radiomics	J. Lipková et al.	2019	Radiother. Oncol.	Growth modelling	Design patient-specific radiotherapy using integrated models	Retrospective GBM cohort; exact n reported in paper	MRI plus genomic/radiomic data	Mechanistic tumour model and optimization	Not applicable	Showed feasibility of model-informed personalized dose design	Simulated control/survival endpoints	Retrospective simulation; not prospective clinical evidence	Blueprint for advanced scenario comparison	10.1016/j.radonc.2019.05.013	https://doi.org/10.1016/j.radonc.2019.05.013
27	Mechanism-based spatiotemporal modeling of temozolomide response in patient-derived glioma xenografts	D. A. Hormuth II et al.	2021	J. R. Soc. Interface	Treatment simulation	Forecast chemotherapy response with calibrated model	Patient-derived xenografts	Longitudinal MRI	Mechanistic growth/drug-response model	Not applicable	Prospective-style calibration/forecasting in preclinical models	Voxel/volume forecast error	Preclinical xenografts, not patients	Supports honest educational chemo scenarios	10.1098/rsif.2020.0941	https://doi.org/10.1098/rsif.2020.0941

28	Repurposing the Public BraTS Dataset for Postoperative Brain Tumour Treatment Response Monitoring	P. J. Sørensen et al.	2024	Tomography	Treatment planning	Test whether preoperative BraTS data can support postoperative response tasks	Public BraTS cases re-analysed	Multiparametric MRI	Dataset audit/re-annotation feasibility study	Not applicable	Shows opportunities and label limitations for postoperative monitoring	Segmentation/availability analyses	BraTS was not designed as longitudinal treatment-response cohort	Guards against misuse of public data	10.3390/tomography10090105	https://doi.org/10.3390/tomography10090105
29	Digital twins for predictive oncology will be a paradigm shift for precision cancer care	A. S. S. G. Ghaffarizadeh et al.	2024	Nat. Med.	Treatment planning	Define predictive oncology digital-twin requirements	Perspective grounded in oncology modelling literature	Multimodal	Digital-twin conceptual framework	Not applicable	Emphasizes iterative data assimilation, uncertainty and validation	Not applicable	Perspective, not validated brain-tumour system	Frames future clinical-grade architecture	10.1038/s41591-024-02816-6	https://doi.org/10.1038/s41591-024-02816-6
30	SLIMBRAIN: Augmented reality real-time acquisition and processing system for hyperspectral classification mapping with depth information for in-vivo surgical procedures	J. Sancho et al.	2023	J. Syst. Archit.	Treatment planning	Overlay real-time tumour-tissue classification during surgery	In-vivo brain-tumour resections; exact n not reported in abstract	Hyperspectral imaging, RGB-D/LiDAR	Real-time classification, point cloud and AR overlay	LiDAR/RGB camera; custom AR	Operated at 14 fps and was verified in surgical procedures	Frame rate; classification/visual verification	Specialized hardware; limited generalizability	Shows real-time intraoperative overlay feasibility	10.1016/j.sysarc.2023.102893	https://doi.org/10.1016/j.sysarc.2023.102893

4. Thematic Literature Review

4.1 AR, VR, and Mixed Reality in Medical Visualization

Across the six visualization papers, the most defensible benefit is improved access to spatial information. Abhari et al. measured training and planning performance rather than assuming that immersion itself was beneficial [1]. Meola et al. found that neurosurgical AR implementations differed in real-data source, tracking, registration, processing, display and perception location [2]. This taxonomy remains useful: a mobile AR educational model, an optical see-through HMD and a microscope head-up display have different error chains and should not share one undifferentiated accuracy claim.

Clinical evidence is strongest where the system is evaluated during real workflow. Roethe et al. prospectively randomized 55 intracranial lesions and found that integrated AR could reduce pointer-based navigation effort, yet virtual content sometimes blocked important anatomy and depth/accuracy ratings remained inferior to conventional neuronavigation [4]. Ivan et al. showed feasible HoloLens incision planning in 11 cases, but a pilot cannot establish outcome benefit [3]. Birlo et al.'s systematic review similarly concluded that much of the OST-HMD literature remained phantom, setup or feasibility work [5].

HoloBrain is particularly relevant because smartphone AR reduces cost and avoids dependence on a headset [6]. A mobile pipeline can use ARKit/ARCore through Unity AR Foundation, but the selected evidence supports the concept of mobile AR more directly than any specific engine choice. Unity is attractive for mesh import, touch gestures and cross-platform deployment; 3D Slicer, VTK and Blender are suitable for segmentation inspection, surface extraction and mesh repair. These are proposed engineering choices, not claims that one selected study validated every tool.

Marker-based registration is simple and reproducible for a classroom tabletop: a printed fiducial provides a known coordinate frame. Markerless plane tracking is more natural but accumulates drift and does not provide anatomical registration. Patient-registered surgical AR requires a separate tracking and registration problem, including fiducials or surface matching and compensation for brain shift. Therefore the student prototype should render a free-standing educational model, not claim surgical navigation. User studies should measure spatial-learning gain, task completion, System Usability Scale, workload and discomfort. Motion sickness is more prominent in VR, while HMD AR adds eyestrain, field-of-view and focal-depth concerns [4], [5].

4.2 Brain MRI Segmentation and 3D Reconstruction

Segmentation is the most technically mature component. The desired labels are commonly whole tumour, tumour core and enhancing tumour, although label definitions vary by BraTS edition and postoperative task. U-Net introduced an encoder–decoder with skip connections that preserves localization while learning contextual features [7]. DeepMedic added multiscale 3D context [9], Myronenko coupled segmentation with autoencoder regularization [10], and nnU-Net automated configuration of preprocessing, patch size, architecture and training strategy [11]. Ranjbarzadeh et al. added attention in a multimodal brain-tumour pipeline [12].

The MRI pipeline should include DICOM-to-NIfTI conversion where needed, de-identification, orientation checks, rigid or affine co-registration, common resampling, skull stripping when appropriate, N4-type bias-field correction and intensity normalization. Preprocessing must be fitted within each training fold to prevent leakage. Missing modalities require either exclusion, modality-robust training or explicit synthesis validation; silently filling missing sequences is unsafe.

Bakas et al. linked expert labels and radiomic features to TCGA glioma imaging, creating a traceable bridge between segmentation, imaging phenotypes and molecular data [8]. However, challenge performance does not guarantee clinical generalization. Multi-institutional variation can help robustness, but postoperative cavities, blood products, treatment effect and acquisition differences create new domains. Dice is useful for overlap, while Hausdorff distance exposes boundary outliers; both should be reported for each tumour region.

For visualization, a predicted voxel mask should be inspected, cleaned only with documented morphology, converted to an isosurface (for example marching cubes), smoothed conservatively, decimated, checked for topology and exported as glTF/GLB or FBX. The transform from voxel indices to physical coordinates must be preserved. Brain regions, tumour subregions and vessels should be separate meshes with consistent coordinate

systems, colour legends and adjustable opacity. Volume rendering may retain uncertainty better than a hard surface; the app can offer both.

4.3 Radiomics and Imaging Biomarkers

Radiomics converts an image and region of interest into numerical descriptors. Shape features quantify volume, surface area, sphericity and elongation; first-order features summarize intensity distributions; texture matrices include GLCM, GLRLM, GLSZM, NGTDM and GLDM; filtered images can yield wavelet features. Aerts et al. established the broader concept of imaging phenotypes and emphasized stability [13]. PyRadiomics subsequently supplied a transparent, open implementation with documented feature definitions [14].

Glioblastoma studies demonstrate both promise and fragility. Kickingereder et al. developed and validated an MRI radiomic survival signature beyond established clinical/radiologic models [15]. He et al. extracted 107 features per sequence and used LASSO-based models to predict IDH, MGMT and TERT status in 81 patients [17]. Carré et al. showed that scanner/protocol standardization and intensity normalization materially affect robustness and classification [16].

A valid pipeline segments first, extracts features with fixed settings, removes unstable or redundant variables, performs selection inside nested cross-validation, and evaluates on a held-out site. Harmonization such as ComBat may reduce centre effects but must not use test-centre information during fitting. Radiomics depends on contour quality, voxel size, intensity normalization and discretization. Small samples combined with thousands of features invite overfitting; leakage occurs when feature selection precedes data splitting. External validation, calibration and decision-curve analysis are therefore more informative than a high internal AUC alone.

For the proposed system, radiomics should be explanatory: show tumour volume, surface-to-volume ratio and a small validated subset of texture features with plain-language definitions. Biomarker and survival outputs should be optional research demonstrations. A feature dashboard is more defensible than presenting an unvalidated “tumour aggressiveness score.”

4.4 Machine Learning for Tumour Progression and Treatment Response

The prediction papers address non-equivalent endpoints. Lao et al. predicted overall survival risk from deep radiomic features [18]. Charlton et al. compared one-year survival models and found that interpretable methods could remain competitive when clinical variables were informative [20]. Kwiatkowska-Miernik et al. estimated progression-free survival in only 51 recurrent high-grade glioma patients [21]. These are risk/time models; none yields a validated future 3D tumour boundary.

Pei et al. did attempt future-segmentation prediction using longitudinal MRI and a reaction–diffusion-derived cell-density feature, but evaluated only nine longitudinal subjects [19]. The study is directly relevant and evidentially limited. Chang et al. classified recurrent glioblastoma versus pseudoprogression in a multi-institutional setting [22]; this is treatment-state discrimination, not growth forecasting. Pseudoprogression can mimic true progression after therapy, so a simulator must avoid equating increased enhancement with biological growth.

Random forests, support-vector machines and gradient boosting are suitable for tabular radiomics/clinical data; CNNs learn spatial features; recurrent networks or temporal transformers may model repeated scans; survival models handle censoring. Multimodal learning can combine imaging, age, performance status, treatment and molecular variables, but missingness and temporal leakage must be controlled. Explainability should match the model: coefficients or SHAP values for tabular models, saliency with sanity checks for imaging, and explicit parameter sensitivity for mechanistic simulation.

Calibration and uncertainty are essential. Classification should report AUC, sensitivity, specificity and calibration; survival should report concordance and time-dependent calibration; future masks should report spatial Dice, surface distance and volume error at specified horizons. Ensembles, Bayesian approximations or conformal prediction can quantify uncertainty, while out-of-distribution detection should warn when scanner, tumour type or postoperative appearance differs from training data. A visually plausible generated scan is not evidence of biological accuracy.

4.5 Mathematical and Computational Tumour-Growth Models

Reaction–diffusion models represent tumour-cell density c(x,t) through diffusion-like invasion and proliferation, often written $\partial c/\partial t = \nabla\cdot(D\nabla c)+\rho c(1-c/K)$. Swanson et al. formalized differential motility in grey and white matter [23], and Harpold et al. reviewed the evolution and clinical interpretation of proliferation–invasion modelling [24]. Parameters have physical meaning, which aids interpretation, but they are only partially identifiable from sparse MRI thresholds.

Patient-specific models estimate proliferation, invasion and sometimes treatment sensitivity from serial scans. Rockne et al. added hypoxia-modulated radiation resistance informed by FMISO-PET [25]. Lipková et al. integrated mechanistic modelling, genomics and radiomics for personalized radiotherapy design [26]. These studies demonstrate hypothesis generation and treatment-design research, not a licensed clinical forecaster.

Cellular automata discretize tissue into local states; agent-based models represent individual cells and interactions; biomechanical models add mass effect and tissue deformation. Each increases parameter burden. Boundary conditions at skull, ventricles and tissue interfaces alter simulated spread; anisotropic diffusion can follow white-matter tracts. Longitudinal calibration is necessary, and validation must compare forecasts made before the future scan—not merely fit all time points retrospectively.

For education, a reaction–diffusion model is valuable because users can see how proliferation and invasion change spatial patterns. For research demonstrations it can support sensitivity analysis and hypothesis generation. Individual clinical forecasting remains uncertain because routine MRI does not directly measure cell density, biology changes under treatment, and model-form error is substantial. The interface should therefore display parameter ranges and ensembles rather than one deterministic surface.

4.6 Treatment Simulation and Patient-Specific Modelling

Treatment visualization and response prediction must be labelled separately. Hormuth et al. calibrated mechanism-based temozolomide-response models in patient-derived xenografts [27]; this is stronger than a purely hypothetical animation but remains preclinical. Sørensen et al. showed why public preoperative BraTS data cannot automatically be treated as a complete postoperative response cohort [28]. A surgery mode can visualize removal of a selected mask and residual-volume calculation, but without deformation, infiltration and postoperative imaging it is a resection demonstration, not outcome prediction.

Radiotherapy simulation can display a prescribed dose field, organs at risk and a modelled cell-kill scenario. Rockne and Lipková provide mechanistic precedents [25], [26], yet real patient planning requires calibrated dose, fractionation, registration, contour review and clinical quality assurance. Chemotherapy scenarios should vary sensitive/resistant parameters rather than claim a certain shrinkage. “No treatment” may show natural-history model ensembles but needs strong warnings.

Digital twins imply repeated assimilation of patient observations into a model that represents current state, forecasts alternatives and updates as new data arrive [29]. The selected paper is a perspective, so it supports design principles—not proof that a brain-tumour digital twin is clinically available. SLIMBRAIN demonstrates a different form of patient-specific assistance: real-time hyperspectral classification mapped into an AR scene at 14 frames per second [30]. It supports feasibility of real-time overlay, not longitudinal response prediction.

A credible scenario comparator should show actual baseline scan, segmented anatomy, model choice, assumptions, uncertainty band and outcome measures such as volume trajectory. It should distinguish real observations from simulated frames by colour and labelling. Clinician review is mandatory whenever patient data are used.

5. Comparative Analysis

Direct numerical ranking across the corpus would be misleading. Segmentation papers differ in BraTS edition, training size, label definition and lesion-wise versus voxel-wise metrics [9]–[12]. Radiomics studies differ in endpoint, feature preprocessing and validation [15]–[17]. Survival AUC, concordance index and progression-free survival error are not interchangeable [18], [20], [21]. AR task time and registration error address usability and geometry rather than biological accuracy [1], [3]–[6]. Accordingly, comparisons below emphasize evidence and task suitability rather than a league table.

- Objective and tumour type: AR papers mainly address cranial lesions or brain-tumour planning; segmentation papers concentrate on adult glioma; prediction studies mix GBM, high-grade glioma and broader brain-tumour cohorts. A model trained on preoperative adult glioma should not be presented as valid for metastases, meningioma, paediatric tumours or postoperative scans.
- Dataset and validation: strongest generalization evidence comes from systematic reviews, multi-institutional resources and external validation [5], [8], [11], [15], [22]. Small pilots and longitudinal cohorts provide feasibility but unstable estimates [3], [19], [21]. Internal cross-validation is not independent clinical validation.
- Accuracy and metrics: Dice and Hausdorff are appropriate for masks; AUC, sensitivity and specificity for binary states; C-index for time-to-event ranking; calibration for probability reliability; registration error and task metrics for AR. Computational requirements rise from tabular models to 3D CNNs, temporal models, physics solvers and real-time rendering. nnU-Net training usually needs a capable GPU, but inference and decimated-mesh rendering can be separated so the mobile device remains responsive.
- Interpretability and real-time suitability: handcrafted radiomics and reaction–diffusion parameters are more directly interpretable than deep latent features, but neither is automatically correct. Mobile AR is real-time after mesh preparation [6]; hyperspectral intraoperative classification reached 14 fps with specialized hardware [30]. Growth simulation may be precomputed into timeline states. A student prototype should prioritize reliable interaction over running every model on-device.

6. Research Gaps

No integrated end-to-end system: AR studies focus on visualization/navigation, whereas progression and treatment papers run offline models; none validates a combined mobile AR, radiomics, growth and scenario platform [3]–[6], [19], [25]–[30].

Longitudinal scarcity: Future-mask work used nine longitudinal subjects, and public BraTS was not designed as a complete postoperative response cohort [19], [28].

Small and single-centre cohorts: Biomarker and PFS studies used 81 and 51 patients, respectively, limiting stable feature selection and calibration [17], [21].

External validation and reproducibility: Many pipelines rely on internal splits; scanner/protocol effects change radiomic features and challenge models may not transfer to routine care [11], [12], [16], [17].

Domain shift: Institution, field strength, sequence parameters, tumour type and treatment state alter input distributions [8], [16], [28].

Uncertainty communication: Most visualization studies display surfaces rather than calibrated spatial uncertainty; deterministic overlays may appear more certain than source segmentation or registration permits [4], [5].

Limited usability evidence: Clinical AR trials are few; occlusion, depth, comfort and workload require broader clinician, student and patient testing [4], [5].

Registration standardization: HMD and mobile systems use differing landmarks, tracking and error definitions, preventing straightforward comparison [2]–[6].

Weak multimodal integration: Radiomics, clinical, molecular and mechanistic variables are usually studied separately; integrated approaches are uncommon and at risk of missing-data bias [17], [20], [26].

Few causal treatment simulations: PFS prediction and recurrence classification do not estimate counterfactual benefit of surgery, radiation or chemotherapy [21], [22], [25]–[27].

Explainability and clinician comparison: Feature importance alone does not establish clinical reasoning; few studies compare calibrated model decisions against clinician judgement across centres [17], [20], [21].

Ethics, privacy and ordinary-device deployment: Patient-linked imaging requires de-identification and governance, while low-cost smartphone evidence remains feasibility-stage [6]. Risk is amplified when simulated growth is communicated to patients as certain.

7. Proposed Research Framework

The architecture should be modular so an error or unsupported claim in one component does not contaminate the rest. The data layer receives anonymized DICOM or NIfTI; validates orientation, spacing and modality; converts when necessary; and stores provenance. The imaging layer registers and normalizes sequences, performs segmentation and produces tumour volumes and 3D surfaces. The analytics layer extracts radiomics, joins selected clinical variables and runs either a validated risk model or a mechanistic scenario engine. The presentation layer renders anatomy in AR/VR with a timeline, labels, measurements, uncertainty and a clear "observed versus simulated" state.

- MRI acquisition and ingestion: local de-identified NIfTI for the student version; governed DICOM gateway only with hospital approval.
- Preprocessing: SimpleITK/NiBabel for orientation, registration, resampling, bias correction and normalization; log transforms and versions.
- Segmentation: MONAI/PyTorch implementation or nnU-Net baseline for whole tumour, core and enhancing tumour; retain probability maps.
- 3D generation: 3D Slicer/VTK marching cubes, conservative smoothing/decimation, anatomical labels and physical-coordinate validation.
- Radiomics: PyRadiomics with fixed binning and resampling; feature selection only inside training folds; optional ComBat fitted without leakage.
- Clinical integration: age, diagnosis, molecular markers and treatment data in a minimal schema with missingness indicators and access control.
- Progression module: choose one of (a) predefined educational trajectories, (b) calibrated reaction–diffusion ensembles, or (c) research-only longitudinal future-mask prediction. Never merge outputs without labels.
- Treatment comparator: surgery as geometric resection demonstration; radiotherapy as dose/response scenario; chemotherapy as parameterized sensitivity scenario; no-treatment as modelled natural-history range.
- Uncertainty/XAI: segmentation probability, scenario ensemble, feature importance, calibration plot, out-of-distribution warning and model card.
- AR/VR rendering: Unity plus AR Foundation for ARKit/ARCore; optional Meta Quest/HTC Vive VR. Touch/controller rotation, scale, clipping planes, opacity, region isolation and reset.
- Timeline: time-stamped observed scans and visually distinct simulated states; tumour-volume curve with confidence band.
- Validation/privacy: expert annotation comparison, longitudinal holdout, site holdout, audit logs, encryption, anonymization and clinician sign-off.

7.1 Minimum Viable Student Prototype

Use public preoperative BraTS/TCIA NIfTI, a pretrained or locally trained nnU-Net/MONAI segmentation model, 3D Slicer or VTK mesh conversion, Unity AR Foundation and local on-device or LAN storage. Provide sample cases, anatomical labels, volume measurement, clipping planes and a timeline containing predefined educational growth scenarios. Allow treatment toggles only as clearly labelled animations based on user-controlled assumptions. Target 30–60 fps after mesh loading. No patient diagnosis, no automatic treatment recommendation, and no claim of future accuracy.

7.2 Advanced Research Version

Add longitudinal MRI, PyRadiomics, clinical/molecular covariates, calibrated survival or recurrence models, reaction–diffusion ensembles, uncertainty maps, prospective forecast locking and comparison with later scans. Containerize preprocessing, track model/data versions and evaluate on a held-out institution. Provide researcher controls for parameter priors and sensitivity analysis, but keep patient-facing language conservative.

7.3 Possible Future Clinical-Grade Version

A clinical-grade version would require hospital DICOM integration, validated device calibration, formal software lifecycle controls, cybersecurity, usability engineering, prospective multicentre trials, documented risk management, regulatory classification, post-market monitoring and accountable clinician oversight. A digital

twin would need repeated data assimilation and uncertainty-aware counterfactual evaluation [29]. These are future requirements, not deliverables of the student prototype.

8. Recommended Datasets

Data set	Tumour	Cases	Modalities	Masks	Longitudinal	Outcome/molecular	Access/license	Official link	Use
BraTS	Adult glioma (editions also include other tasks)	Varies by year; hundreds to >2,000 across modern task families	T1, T1CE, T2, FLAIR	Expert segmentation masks	Limited in classic preoperative releases	Some survival/clinical labels by edition	Challenge agreement/data-use terms	https://www.synapse.org/brats	Segmentation benchmarking and sample meshes
TCGA-GBM via TCIA	Glioblastoma	262 subjects listed by TCIA collection; usable modality counts vary	MRI, CT and other studies by subject	Expert labels available for a curated subset [8]	Some repeated studies; not a standardized longitudinal cohort	Linked TCGA genomics/clinical data subject to portals	TCIA data-use policy; some linked data controlled	https://www.cancerimagingarchive.net/collection/tcga-gbm/	Radiogenomics and retrospective examples
TCGA-LGG via TCIA	Lower-grade glioma	199 subjects listed by TCIA collection; usable counts vary	MRI	Curated labels available for a subset [8]	Not consistently longitudinal	Linked TCGA molecular/clinical data	TCIA/TCGA access terms	https://www.cancerimagingarchive.net/collection/tcga-lgg/	Subtype/radiomics studies
UPENN-GBM	Glioblastoma	630 de-identified subjects in the release description	Multiparametric MRI; clinical and molecular variables	Automated and expert-revised annotations provided by collection	Longitudinal studies available for subsets; verify task eligibility	Clinical, genomic and radiomic information available by release	TCIA license/data-use terms	https://www.cancerimagingarchive.net/collection/upenn-gbm/	External testing, radiomics and prognosis research
ACRIN 6684 / RTOG 0825	Newly diagnosed GBM clinical trial imaging	621 subjects listed by TCIA collection; image availability varies	MRI	Not uniformly supplied as research masks	Serial trial imaging available	Treatment/outcome data under access conditions	Some clinical data require application	https://www.cancerimagingarchive.net/collection/acrin-6684/	Treatment-response and longitudinal research

Counts above describe collection-level releases and may change as collections are corrected or expanded. Before analysis, the team must verify the current official collection page, subject-level availability, license and whether the required labels or outcomes are actually downloadable. Dataset existence does not imply that every subject has every modality or endpoint.

9. Evaluation Framework

Segmentation

Report Dice similarity coefficient, intersection over union, sensitivity, specificity, precision, 95th-percentile Hausdorff distance and average surface distance separately for whole tumour, tumour core and enhancing tumour. Use patient-level confidence intervals, expert annotations, failure-case review and an external-site set.

Progression prediction

For probability models: AUC, sensitivity/specificity at a prespecified threshold, Brier score, calibration slope/intercept and expected calibration error. For survival: concordance index, integrated Brier score and time-dependent calibration. For future masks: volume MAE, spatial Dice, surface distance and longitudinal horizon-specific validation. Lock the prediction before viewing the future scan.

AR/VR

Measure fiducial and target registration error, drift, frame rate, motion-to-photon/rendering latency, load time, task completion, localization error, System Usability Scale, NASA-TLX, satisfaction, presence/immersion, simulator sickness and pre/post learning improvement. Compare against 2D MRI or a standard 3D viewer.

Safety and reliability

Display confidence intervals and spatial uncertainty; test out-of-distribution detection, missing modalities, corrupted headers and segmentation failures; show warnings; require human approval; and compare with independent expert annotations. Define stop conditions when registration or model confidence is unacceptable.

10. Ethical and Clinical Considerations

- Privacy: Remove direct DICOM identifiers, inspect burned-in pixels and private tags, use coded subject IDs, minimize retained metadata, encrypt data in transit and at rest, separate identity keys, apply role-based access and record audit events. Public datasets still require compliance with their licences.
- Consent and governance: Hospital data require ethics approval or an applicable waiver, data-use agreements and clear consent for education or patient counselling. Secondary use and model training must be disclosed where required.
- Bias: Dataset composition, scanner availability, tumour subtype and treatment access influence model performance. Report subgroup results and avoid assuming that BraTS represents every hospital population.
- Explainability and uncertainty: Users must see whether a frame is measured, segmented, predicted or simulated. Parameter ranges, probability calibration and model limitations should accompany animations. Synthetic images need watermarks or equivalent provenance to prevent confusion with acquired scans.
- Clinical responsibility and regulation: A system that influences diagnosis or treatment may qualify as medical-device software and would require jurisdiction-specific regulatory evidence. Clinicians retain responsibility; a classroom prototype must not be used to choose a patient's therapy.
- Patient communication: Immersive graphics can appear authoritative and may increase distress. Counselling should use neutral language, avoid deterministic timelines, allow clinicians to hide experimental features and provide a plain-language explanation that scenarios are not medical predictions.

11. Conclusion

Current research supports an integrated educational prototype, but the components have unequal maturity. Multimodal MRI segmentation, mesh generation and interactive 3D visualization are the most mature. Radiomics can add quantitative descriptors when preprocessing and validation are controlled. AR can improve spatial understanding, yet registration, occlusion, depth perception and usability remain active engineering problems. Survival, recurrence and pseudoprogression models answer clinically relevant questions but do not simulate spatial growth. Reaction–diffusion and treatment models provide interpretable research scenarios, not certain patient forecasts. The recommended student direction is therefore a mobile AR brain/tumour viewer with validated segmentation, transparent volume and radiomic measurements, predefined educational scenarios and an optional parameterized growth demonstration with uncertainty. Clinical deployment would require longitudinal multicentre data, prospective validation, human-factors engineering, cybersecurity, governance and regulatory evidence.

12. Future Scope

Future research may develop iterative digital twins that assimilate new scans; multimodal foundation models combining images, reports, genomics and treatment; temporal transformers for longitudinal forecasting; federated learning across hospitals; multicentre domain-generalization studies; physics-informed neural networks that combine reaction–diffusion constraints with data; uncertainty-aware spatial ensembles; personalized radiotherapy optimization; real-time surgical AR that compensates for brain shift; haptic tumour-resection training; collaborative VR for remote planning; and clinician-in-the-loop systems that record corrections and recalibrate safely. Each direction needs prospective validation and explicit governance before clinical influence [25], [26], [29], [30].

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15. Citation-Verification Appendix

Ref.	Paper title	First author	Year	DOI	PMID	Publisher/database	Peer reviewed	OA status	Type	Verification
1	Training for Planning Tumour Resection: Augmented Reality and Human Factors	Abhari	2015	10.1109/TBME.2014.2385874	Not reported	IEEE Trans. Biomed. Eng.	Yes	Verify via DOI/publisher	Original research	Fully verified
2	Augmented reality in neurosurgery: a systematic review	Meola	2017	10.1007/s10143-016-0732-9	Not reported	Neurosurg. Rev.	Yes	Verify via DOI/publisher	Review/perspective	Fully verified
3	Augmented reality head-mounted display-based incision planning in cranial neurosurgery: a prospective pilot study	Ivan	2021	10.3171/2021.5.FOCUS20735	Not reported	Neurosurg. Focus	Yes	Verify via DOI/publisher	Original research	Fully verified
4	Augmented reality visualization in brain lesions: a prospective randomized controlled evaluation of its potential and current limitations in navigated microneurosurgery	Roethe	2022	10.1007/s00701-021-05047-x	Not reported	Acta Neurochir.	Yes	Verify via DOI/publisher	Original research	Fully verified
5	Utility of optical see-through head mounted displays in	Birlo	2022	10.1016/j.media.2022.102361	Not reported	Med. Image Anal.	Yes	Verify via DOI/publisher	Review/perspective	Fully verified

	augmented reality-assisted surgery: a systematic review									
6	HoloBrain: 3D low-cost mobile augmented reality rendering of brain tumour using the GVF snake model segmentation	Guerroudji	2024	10.1080/21681163.2024.2302051	Not reported	Comput. Methods Biomech. Biomed. Eng. Imaging Vis.	Yes	Verify via DOI/publisher	Original research	Fully verified
7	U-Net: Convolutional Networks for Biomedical Image Segmentation	Ronneberger	2015	10.1007/978-3-319-24574-4_28	Not reported	MICCAI	Yes	Verify via DOI/publisher	Data/method paper	Fully verified
8	Advancing The Cancer Genome Atlas glioma MRI collections with expert segmentation labels and radiomic features	Bakas	2017	10.1038/sdata.2017.117	Not reported	Sci. Data	Yes	Verify via DOI/publisher	Data/method paper	Fully verified
9	Brain tumor segmentation with DeepMedic	Kamnitsas	2017	10.1016/j.media.2016.10.004	Not reported	Med. Image Anal.	Yes	Verify via DOI/publisher	Original research	Fully verified
10	3D MRI brain tumor segmentation using autoencoder regularization	Myronenko	2019	10.1007/978-3-030-11726-9_28	Not reported	BrainLes/MICCAI	Yes	Verify via DOI/publisher	Original research	Fully verified
11	nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation	Isensee	2021	10.1038/s41592-020-01008-z	Not reported	Nat. Methods	Yes	Verify via DOI/publisher	Data/method paper	Fully verified
12	Brain tumor segmentation based on deep learning and an attention mechanism using MRI multi-modalities brain images	Ranjbarzadeh	2021	10.1038/s41598-021-90428-8	Not reported	Sci. Rep.	Yes	Verify via DOI/publisher	Original research	Fully verified
13	Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach	Aerts	2014	10.1038/ncomms5006	Not reported	Nat. Commun.	Yes	Verify via DOI/publisher	Original research	Fully verified
14	Computational radiomics system to decode the radiographic phenotype	Griethuysen	2017	10.1158/0008-5472.CAN-17-0339	Not reported	Cancer Res.	Yes	Verify via DOI/publisher	Data/method paper	Fully verified
15	Radiomic profiling of glioblastoma: identifying an imaging predictor of patient survival with improved performance over established clinical and radiologic risk models	Kickingereder	2016	10.1148/radiol.2016160845	Not reported	Radiology	Yes	Verify via DOI/publisher	Original research	Fully verified
16	Standardization of brain MR images across machines and protocols: bridging the gap for MRI-based radiomics	Carré	2020	10.1038/s41598-020-69298-z	Not reported	Sci. Rep.	Yes	Verify via DOI/publisher	Original research	Fully verified
17	Multiparametric MR radiomics in brain glioma: models	He	2022	10.1186/s12880-022-00865-8	Not reported	BMC Med. Imaging	Yes	Verify via DOI/publisher	Original research	Fully verified

	comparation to predict biomarker status									
18	A deep learning-based radiomics model for prediction of survival in glioblastoma multiforme	Lao	2017	10.1038/s41598-017-10649-8	Not reported	Sci. Rep.	Yes	Verify via DOI/publisher	Original research	Fully verified
19	Longitudinal brain tumor segmentation prediction in MRI using feature and label fusion	Pei	2020	10.1016/j.bspc.2019.101648	Not reported	Biomed. Signal Process. Control	Yes	Verify via DOI/publisher	Original research	Fully verified
20	Development of prediction models for one-year brain tumour survival using machine learning: a comparison of accuracy and interpretability	Charlton	2023	10.1016/j.cmpb.2023.107482	Not reported	Comput. Methods Programs Biomed.	Yes	Verify via DOI/publisher	Original research	Fully verified
21	Estimating Progression-Free Survival in Patients with Primary High-Grade Glioma Using Machine Learning	Kwiatkowska-Miernik	2024	10.3390/jcm13206172	Not reported	J. Clin. Med.	Yes	Verify via DOI/publisher	Original research	Fully verified
22	Prediction of Pseudoprogression versus Progression using Machine Learning Algorithm in Glioblastoma	Jang	2018	10.1038/s41598-018-31007-2	Not reported	Sci. Rep.	Yes	Verify via DOI/publisher	Original research	Partially verified
23	A quantitative model for differential motility of gliomas in grey and white matter	Swanson	2000	10.1046/j.1365-2184.2000.00177.x	Not reported	Cell Prolif.	Yes	Verify via DOI/publisher	Original research	Fully verified
24	The evolution of mathematical modeling of glioma proliferation and invasion	Harpold	2007	10.1097/01.jnen.0000245080.64495.7e	Not reported	J. Neuropathol. Exp. Neurol.	Yes	Verify via DOI/publisher	Review/perspective	Fully verified
25	A patient-specific computational model of hypoxia-modulated radiation resistance in glioblastoma using 18F-FMISO-PET	Rockne	2015	10.1098/rsif.2014.1174	Not reported	J. R. Soc. Interface	Yes	Verify via DOI/publisher	Original research	Fully verified
26	Personalized radiotherapy design for glioblastoma using mathematical tumor models, genomics and radiomics	Lipková	2019	10.1016/j.radonc.2019.05.013	Not reported	Radiother. Oncol.	Yes	Verify via DOI/publisher	Original research	Fully verified
27	Mechanism-based spatiotemporal modeling of temozolomide response in patient-derived glioma xenografts	Il	2021	10.1098/rsif.2020.0941	Not reported	J. R. Soc. Interface	Yes	Verify via DOI/publisher	Original research	Fully verified
28	Repurposing the Public BraTS Dataset for Postoperative Brain Tumour Treatment Response Monitoring	Sørensen	2024	10.3390/tomography10090105	Not reported	Tomography	Yes	Verify via DOI/publisher	Original research	Fully verified
29	Digital twins for predictive oncology will	Ghaffarizadeh	2024	10.1038/s41591-024-02816-6	Not reported	Nat. Med.	Yes	Verify via DOI/publisher	Review/perspective	Partially verified

	be a paradigm shift for precision cancer care									
30	SLIMBRAIN: Augmented reality real-time acquisition and processing system for hyperspectral classification mapping with depth information for in-vivo surgical procedures	Sancho	2023	10.1016/j.sysarc.2023.102893	Not reported	J. Syst. Archit.	Yes	Verify via DOI/publisher	Original research	Fully verified

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